HANYU YANG

CRM data analytics / Software systems / Applied AI

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Profile

Software engineering graduate and postgraduate student in AI, Ethics and Society at the University of Hong Kong. Experience spans CRM data analytics, AI workflow development, full-stack software, multimodal research, and distributed data systems.

01 / Professional experience

Professional experience

HSBC — **Mar 2026 - Jul 2026**

CRM Data Analyst Intern · Wealth and Personal Banking — Guangzhou, China

- Integrated customer data from core banking, credit-card, and mobile channels with SQL and Python on GCP BigQuery and Data Science Workbench, building a unified customer view and tagging system for precise WPB client profiling.
- Performed data cleaning, exploratory analysis, and predictive modeling in Python to support customer segmentation and evidence-led marketing campaign decisions.
- Contributed to precision-marketing strategy and campaign deployment through SmartCRM and the PEGA decisioning platform, supporting omni-channel customer lifecycle management.
- Supported omnichannel customer-communications delivery by coordinating handover requirements, UAT inputs, execution readiness, channel schedules, and stakeholder sign-offs across in-app, eDM, and marketing-push journeys.

Tools: SQL / Python / GCP BigQuery / Data Science Workbench / SmartCRM / PEGA / UAT

Guangzhou Xin'an Data Co., Ltd. — **Jul 2025 - Aug 2025**

AI Development Intern · Digital Innovation Department — Guangzhou, China

- Designed and deployed a full-stack AI prompt-optimization platform with authentication, community libraries, and knowledge-sharing capabilities for more than 100 internal users.
- Integrated Model Context Protocol with Dify to automate AI-driven reporting workflows.
- Developed RESTful APIs for third-party integration and enterprise data exchange.

Tools: MCP / Dify / REST API / Full-stack / AI workflows

02 / Education

Education

The University of Hong Kong — **Sep 2026 - Aug 2027**

Master of Arts in AI, Ethics and Society — Hong Kong SAR

- Postgraduate study connecting AI systems with their ethical and societal contexts.

Guangdong University of Foreign Studies · School of Information Science and Technology — **Sep 2022 - Jun 2026**

Bachelor of Science in Software Engineering — Guangzhou, China

- GPA: 88/100 · Top 10%.
- Selected modules: Text Information Processing (99), Network Programming (96), Natural Language Processing (96), Software Architecture (96), Big Data Processing Technology (94), Further Mathematics (91), and Web Design (90).

03 / Research experience

Research experience

Guangdong University of Foreign Studies — **Oct 2024 - Dec 2024**

Research Contributor · Fake News Detection through Multimodal Attention

Guangzhou, China

- Proposed a semantic-conflict recognition and multimodal-attention framework for image-text inconsistency.
- Added KL-divergence and MSE modules to CLIP alignment for quantifying cross-modal semantic discrepancies.
- Combined BERT, ResNet-50, and cross-modal self-attention, reaching 88.9% accuracy and improving on the CLIP baseline by 2.8%.

Tools: BERT / ResNet-50 / CLIP / Multimodal attention

University of Cambridge — **Aug 2024**

Research Leader · Skin Cancer Classification with CNN and Attention

Cambridge, UK

- Led development of a lightweight enhanced-AlexNet CNN for benign-versus-malignant dermoscopic lesion classification.
- Integrated spatial attention to focus the model on diagnostically relevant regions.
- Improved classification accuracy by 15% through preprocessing optimization and background-noise suppression.

Tools: CNN / AlexNet / Spatial attention / Medical imaging

Guangdong University of Foreign Studies — **Oct 2023 - Nov 2023**

Algorithm Designer · Autonomous Parking Detection

Guangzhou, China

- Redesigned YOLOv5s with a GhostNet backbone and CBAM attention for parking-slot detection under occlusion and variable lighting.
- Trained in PyTorch with RMSProp and Mosaic augmentation for 500 epochs on RTX 3090.
- Reached 97.1% AP and 93.7% AR while reducing inference latency by 20% against the YOLOv5s baseline.

Tools: PyTorch / YOLOv5s / GhostNet / CBAM

04 / Engineering projects

Engineering projects

Guangdong University of Foreign Studies — **Apr 2025 - Jul 2025**

Front-End Developer · International Chinese Education Platform

Guangzhou, China

- Built a responsive Vue 3 student portal with real-time class management and messaging for more than 500 international students.
- Redesigned live-class and chat layouts to improve clarity, responsiveness, and engagement.
- Added lazy loading and virtual scrolling for smooth interaction under real-time data flow.

Tools: Vue 3 / Composition API / Virtual scrolling

Guangdong University of Foreign Studies — **May 2025 - Jun 2025**

Project Leader · YouTube Sentiment Analysis on English Comments

Guangzhou, China

- Collected more than 50,000 English comments across five topics through the YouTube API.
- Built NLTK preprocessing and TF-IDF/Word2Vec feature pipelines, then tuned LinearSVC and XGBoost with GridSearchCV.
- Reported 90.5% validation accuracy and 0.89 F1 for the classical-ML pipeline.
- Deployed a Streamlit prediction interface for more than 100 internal users with average response below two seconds.

Tools: Python / NLTK / LinearSVC / XGBoost / Streamlit

Course engineering project — **Nov 2024 - Dec 2024**

Data Engineer · Global Economic Data Analysis

Guangzhou, China

- Processed 8,519 records with 27 features through Hadoop HDFS on a three-node cluster.
- Built HiveQL aggregation, PySpark GDP analysis across more than 150 countries, and Flink/Kafka streaming pipelines.
- Reduced complex batch-query response time by 40% with Spark and improved time-series query efficiency by 15% through Hive partitioning.

Tools: Hadoop / Hive / Spark / Flink / Kafka

Campus services project **Nov 2024 - Dec 2024**

NLP Developer · ChatGDUFS Campus Q&A

Guangzhou, China

- Built a LangChain RAG pipeline with Chroma and more than 2,000 documentation embeddings.
- Deployed a Flask query API averaging under two seconds for more than 100 daily queries.
- Fine-tuned Qwen-7B with LoRA on 5,000 campus Q&A pairs, improving reported domain accuracy from 72% to 86%.
- Added session-aware multi-turn dialogue across six conversation modes.

Tools: LangChain / RAG / Chroma / Qwen-7B / LoRA / Flask

05 / Technical toolkit

Technical toolkit

Programming	Frameworks & tools	Languages
C++ / Java / Python / SQL / Vue 3 / React / JavaScript / Git	TensorFlow / PyTorch / MySQL / Docker / Redis / Hadoop / Spark / Flink / Kubernetes	English · IELTS 7 / Mandarin · Native / Cantonese · Fluent

06 / Awards & recognition

Awards & recognition

01 China International College Students' Innovation Competition - Outstanding Award, Jul. 2025.

02 14th Challenge Cup Guangdong College Students Entrepreneurship Competition - Bronze Award, Jul. 2024.

03 National College Students English Vocabulary Ability Competition - Third Prize (National), Jun. 2024.

04 Guangdong University of Foreign Studies - Third-Class Scholarship for Outstanding Students, Oct. 2023.

05 Guangdong Youth Science and Technology Innovation Competition - Silver Award, Oct. 2019.

07 / Publication

Publication

Yang, H., Bai, J., He, C., Wang, Y., & Liang, W. (2024). Synthesis, characterization, and testing of rare earth-doped GdBO3 luminescent materials. Guangdong Chemical Industry, 51(10), 50-54.

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